Requested Is Not Realised: Measuring Rate Adherence and Output Expansion in Neural Prompt Compression

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***Abstract*—Prompt compression lowers the cost of large language model inference, but deployed compressors take the compression rate as a hyperparameter supplied by the user and applied uniformly to inputs that tolerate compression very differently. We report a controlled measurement of what a requested rate actually delivers. Using LLMLingua-2 as the compressor and a local instruction-tuned 3B target model, we run a 630-cell grid over 30 instances from three task families and seven compression budgets, plus a separate 1,044-point sweep over 174 instances that requires no target-model calls. Three results follow. First, the realised compression rate is consistently below the requested one when measured in the target model's tokenizer, with a signed error between -0.025 and -0.053 whose bootstrap confidence intervals all exclude zero. The error varies non-monotonically with the requested budget and differs by a factor of about three across task families, so a single scalar offset does not remove it; a calibration table indexed by budget and family removes 79.5% of it on held-out instances, which also bounds what a per-instance model could add. Second, compression increases output length on two of three families, but the increase does not change the cost-optimal budget: across all 25 evaluable instances and output price ratios from 1 to 5, the cheapest safe budget was always the smallest safe budget, because input reduction exceeds output growth by roughly two orders of magnitude in short-answer tasks. Third, per-instance heterogeneity in compression tolerance is visible in the measured curves but is not statistically resolvable at three samples per cell; an extrapolation from the measured noise floor indicates that roughly twenty samples per cell are required. The last result is a constraint on experimental design for this line of work rather than a finding about compression itself. All measurements come from a single target model and a single compressor checkpoint.**

***Index Terms*—prompt compression, large language models, inference cost, measurement, experimental design**

I. Introduction

Prompt compression reduces the number of input tokens sent to a large language model by deleting or rewriting parts of the prompt. Token-level methods such as LLMLingua [1] and its successors [2], [3] now achieve large reductions with modest quality loss, and the compression rate is the primary control the user has over the resulting trade-off.

That control is weaker than it appears, in two ways that this paper measures.

The first concerns what the rate actually means. A compressor is asked for a rate b and returns text whose realised rate r is measured after the fact. If $r \neq b$ systematically, then a reported saving computed from b describes a request rather than a result. Prior work has observed that adherence is imperfect [4]; we characterise its structure across budgets and task families, measured in the tokenizer of the model that is actually billed.

The second concerns what the objective should be. Providers bill input and output tokens separately, and output tokens usually cost more. Compression changes output length as well as input length, and under aggressive truncation that change can be large [5]. If output growth is large enough, then a more aggressive budget can be both quality-safe and more expensive, which would make the common rule of taking the smallest safe budget incorrect. Whether that happens is an empirical question about the output-to-input token ratio in a given workload, and we test it directly.

Our contributions are measurements rather than a method:

1) A characterisation of LLMLingua-2 rate adherence in the target tokenizer, over 1,044 measurements and 174 instances, showing a consistent negative signed error with a non-monotonic dependence on the requested budget and a factor-of-three spread across task families. We then measure, on held-out instances, how much of that error a calibration table removes, which bounds the headroom left for any per-instance corrector.
2) A test of whether compression-induced output expansion changes the cost-optimal budget. Expansion is present, but the optimum does not move in any of the 25 evaluable instances at output price ratios from 1 to 5. We report this bounded negative alongside the positive expansion measurement.
3) A power analysis of per-instance frontier estimation, showing that the sampling noise floor, and not the

number of instances, is the binding constraint, and quantifying the samples per cell required to clear it.

We do not propose a compression method, a rate predictor, or a selection policy, and we make no claim about model families other than the one measured.

II. Related Work

Compression methods. LLMLingua [1] allocates a user-specified budget across prompt segments using a small language model's perplexity. LongLLMLingua [2] adds question-aware coarse-to-fine selection for long contexts. LLMLingua-2 [3] reframes compression as binary token classification with a bidirectional encoder distilled from GPT-4 annotations, and is both faster and more robust out of domain, which is why we use it here. In all three the rate is supplied by the user.

Adaptive rate selection. AdaComp [6] predicts how many retrieved documents to keep, at document granularity in a retrieval-augmented setting. Guo et al. [7] adapt compression to input complexity at embedding granularity. Nagle et al. [8] characterise the fundamental distortion-rate limit of prompt compression with oracle access, which bounds what any selector could achieve rather than providing one. These works motivate per-instance rate selection; none of them reports the adherence gap between requested and realised rates.

Cost and latency in deployment. Kummer et al. [4] present a large-scale study of compression overhead against decoding latency across several open models and three GPU classes, and report end-to-end speedups only inside a matched operating window of prompt length, rate and hardware. Their study includes rate adherence among the quantities measured, so we do not claim priority for observing it; our contribution is the shape of the error as a function of the requested budget and its variation across task families, measured in the target tokenizer. Johnson [5], [9] reports that aggressive compression can raise total cost through output expansion, with a strong benchmark effect (a reported 56× expansion on MBPP against 5× on HumanEval). That work uses truncation as the compression operator and studies code generation. We measure expansion with a neural compressor and task-native metrics, and we find that in short-answer question answering the effect does not reach the threshold needed to move the cost optimum.

Risk control. Distribution-free risk control [10], [11] and its application to prompting decisions [12] offer a route to calibrated rate selection. We cite this line as context for future work; no guarantee is constructed or claimed in this paper.

III. Method

A. Definitions

An instance is a pair (x, q) of context and query. A compressor ψ takes a requested budget $b \in (0, 1]$ and returns compressed context $\psi(x, q, b)$. The realised rate is

$$r(x, q, b) = \frac{|T(\psi(x, q, b))|}{|T(x)|}, \tag{1}$$

where T is the *target model's* tokenizer. Measuring r in the compressor's own tokenizer would report a quantity that does not determine the bill; the two disagree, and the target-side count is the one that is charged. The adherence error is the signed quantity $r - b$.

Quality $\rho(x, b) \in [0, 1]$ is the task metric, estimated by averaging over k stochastic generations. Because ρ is estimated, every per-instance quantity derived from it carries sampling noise, which Section III-C treats explicitly.

B. Cost model

Total cost in token units is

$$C(x, b) = T_{\text{in}}(x, b) + \gamma\, T_{\text{out}}(x, b), \qquad \gamma = c_{\text{out}}/c_{\text{in}}. \tag{2}$$

Local inference has no provider bill, so USD in our ledger is zero by construction. We therefore report cost in token units under an explicit γ and sweep $\gamma \in \{1, \ldots, 5\}$, which states conclusions as a function of the price ratio rather than of one vendor's price list.

Given a tolerance ε, the safe set at instance x is $\mathcal{B}(x) = \{b : \rho(x, b') \geq \rho(x, 1) - \varepsilon \;\; \forall b' \geq b\}$, defined over the suffix so that a single noisy dip does not certify an aggressive budget as safe. The two candidate rules are the smallest safe budget $\min \mathcal{B}(x)$ and the cheapest safe budget $\arg\min_{b \in \mathcal{B}(x)} C(x, b)$. They coincide whenever C is increasing in b; they differ only if output growth outweighs input reduction. We measure how often they differ.

C. Statistics for heterogeneity

The quantity of interest for per-instance rate selection is whether instances differ in compression tolerance by more than estimation noise. The natural label, the smallest safe budget b^*, is a threshold crossing of a noisy curve and inherits the noise of every budget it crosses. We therefore also report a smoother statistic, the compression sensitivity

$$S(x) = \rho(x, 1.0) - \rho(x, 0.2), \tag{3}$$

which uses only the two endpoints and is better conditioned at small k.

For both statistics we estimate a sampling noise floor by split-half resampling: the k generations of each instance are split in two, the statistic is computed on each half, and the mean squared half-difference divided by two estimates the variance attributable to sampling. Heterogeneity is declared only if the bootstrap lower bound on the between-instance variance exceeds twice this floor.

Instances with $\rho(x, 1) = 0$ are excluded from these tests. A prompt the model already fails without compression has no compression tolerance to measure: every budget trivially satisfies the safety condition, so its b^* reflects model failure rather than compression behaviour.

IV. Experimental Setup

Table I summarises the configuration. The three families are drawn from LongBench (MultiFieldQA-en, 2WikiMQA) and GSM8K. MultiFieldQA and 2WikiMQA are scored with token-level F1; GSM8K uses exact match on the extracted

TABLE I
EXPERIMENTAL CONFIGURATION. ALL VALUES ARE RECORDED IN THE RUN MANIFEST COMMITTED WITH THE LEDGER.

Compressor	LLMLingua-2, CPU backend
Checkpoint	`bert-base-multilingual-cased-meetingbank`
Target model	Llama-3.2-3B-Instruct, Q4_K_M, Ollama
Context window	8192 tokens, enforced
Budgets b	1.0, 0.8, 0.65, 0.5, 0.4, 0.3, 0.2
Samples k	3 per cell, temperature 0.7, seeds 0–2
Families	MultiFieldQA (F1), 2WikiMQA (F1), GSM8K (exact match)
Instances	10 per family, 30 total
Grid	630 cells, 0 failures
Adherence sweep	1,044 points, 174 instances, 6 budgets
Context length	500–3500 tokens (selection window)
Tolerance	$\varepsilon = 0.05$

numeric answer, and its context is the few-shot exemplar block, which is the setting LLMLingua-2 itself targets.

Instances were restricted to contexts between 500 and 3500 tokens. Families whose natural length exceeds the target's usable window, specifically HotpotQA (median 14,114 tokens) and QMSum (13,505), were excluded rather than truncated to fit. Truncating them would remove the supporting passage and force $\rho(x, 1) = 0$, producing degenerate labels for a reason unrelated to compression.

Two controls matter for interpretation. The harness raises an error if the server reports that the prompt reached the context ceiling, because silent truncation at $b = 1.0$ would contaminate the reference quality against which every frontier is measured. Token counts are taken from the server's reported usage rather than from local re-tokenisation.

The realised rate is measured with a GPT-2 byte-pair tokenizer as a proxy, since the Llama-3.2 tokenizer is gated. This shifts the absolute level of the adherence error but not its ordering across budgets; the substitution is recorded in the run manifest.

Confidence intervals are bootstrap percentile intervals over instances with 10,000 resamples. The grid consumed 1.28 hours of measured wall time on an Apple M3 with 16 GB of unified memory, of which compression accounted for 23.4%.

V. RESULTS

A. *Which families support quality conclusions*

Table II gives the measured quality curves and Table III the preconditions that determine which of them can be interpreted. Two of the three families are compromised as quality measurements, for different reasons, and we state this before reporting quality results rather than after.

On 2WikiMQA the target model scores 0.105 uncompressed. At that level the model is close to the floor of the metric and the curve is nearly flat across the whole budget range (0.105 down to 0.064), so there is little quality to lose and no informative frontier. On GSM8K, 4 of 10 instances are not solved at $b = 1.0$ in any of the three samples; for those instances every budget trivially satisfies the safety condition,

TABLE II
MEAN QUALITY AGAINST REQUESTED BUDGET, BY FAMILY. MULTIFIELDQA AND 2WIKIMQA ARE TOKEN-F1; GSM8K IS EXACT MATCH. EACH CELL AVERAGES 30 GENERATIONS (10 INSTANCES $\times$ 3 SAMPLES).

b	MultiFieldQA	2WikiMQA	GSM8K
1.00	0.544	0.105	0.467
0.80	0.552	0.101	0.633
0.65	0.534	0.101	0.633
0.50	0.431	0.096	0.567
0.40	0.432	0.072	0.667
0.30	0.377	0.071	0.533
0.20	0.333	0.064	0.667

TABLE III
VALIDITY PRECONDITIONS PER FAMILY. AN INSTANCE IS SOLVABLE IF $\rho(x, 1) > 0$; UNSOLVABLE INSTANCES HAVE NO COMPRESSION TOLERANCE TO MEASURE. ONLY MULTIFIELDQA SATISFIES BOTH PRECONDITIONS.

Family	solvable	$\rho(x, 1)$	status
MultiFieldQA	10/10	0.544	usable
2WikiMQA	9/10	0.105	near-floor accuracy
GSM8K	6/10	0.467	4 degenerate instances

and their b^* records model failure rather than compression tolerance. The GSM8K column of Table II is also visibly non-monotone: its highest values (0.667) occur at $b = 0.4$ and $b = 0.2$ and its lowest (0.467) at $b = 1.0$. At $k = 3$ on a binary metric this is consistent with sampling noise rather than with compression improving accuracy.

MultiFieldQA satisfies both preconditions, and its curve declines monotonically apart from a small rise at $b = 0.8$. Quality conclusions in the remainder of this section therefore rest on MultiFieldQA. The adherence result in Section V-B does not depend on generation at all and is unaffected by these preconditions.

B. *Rate adherence*

LLMLingua-2 returns less text than requested at every budget tested. Table IV gives the pooled measurement and Fig. 1 shows its structure. Two features are relevant to anyone using the requested rate as a control variable.

The error is not constant. It grows from -0.025 at $b = 0.2$ to -0.053 at $b = 0.65$ and then shrinks to -0.040 at $b = 0.8$, an inverted-U shape in the requested budget. A single additive or multiplicative correction therefore cannot remove it.

The error is family-dependent. At $b = 0.65$ the mean signed error is -0.074 on 2WikiMQA, -0.060 on MultiFieldQA and -0.025 on GSM8K. The dependence on input text, and not only on the requested budget, is what would make a learned per-instance correction the appropriate object rather than a lookup table.

The errors are modest in absolute terms, at most about 5 percentage points of the original prompt, and their direction matters. Because $r < b$ the compressor removes *more* text than asked, so the realised saving exceeds the requested one: at $b =$

TABLE IV

REQUESTED AGAINST REALISED COMPRESSION RATE, POOLED OVER THREE FAMILIES ($n = 174$ INSTANCES PER BUDGET). ALL CONFIDENCE INTERVALS EXCLUDE ZERO.

b	realised r	s.d.	$r - b$	95% CI
0.20	0.1751	0.0104	-0.0249	$[-0.0265, -0.0233]$
0.30	0.2656	0.0143	-0.0344	$[-0.0365, -0.0323]$
0.40	0.3581	0.0190	-0.0419	$[-0.0448, -0.0391]$
0.50	0.4492	0.0196	-0.0508	$[-0.0538, -0.0479]$
0.65	0.5973	0.0271	-0.0527	$[-0.0568, -0.0487]$
0.80	0.7596	0.0327	-0.0404	$[-0.0453, -0.0355]$

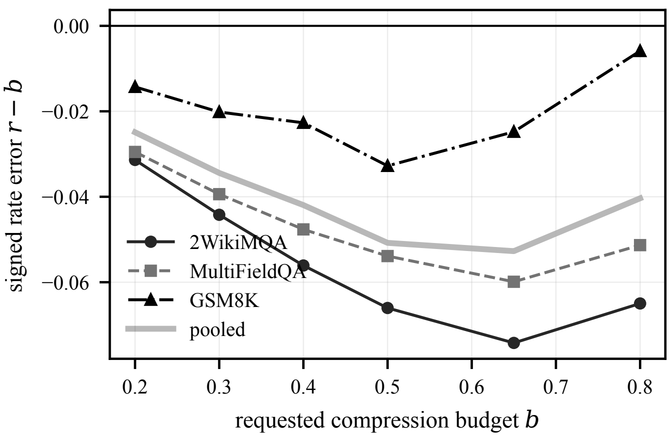

Fig. 1. Signed rate error $r-b$ against requested budget, by family and pooled. The error is negative at every budget, is largest in magnitude near $b = 0.65$, and differs by roughly a factor of three between 2WikiMQA and GSM8K.

0.65 the request implies a 35% reduction and the measurement gives 40.3%. The discrepancy therefore favours the user on cost, and its consequence lies in quality instead. A user who selects b to stay inside a quality tolerance is served at the more aggressive point r, so the delivered quality corresponds to a budget they did not choose, and the effect is largest near $b = 0.65$ where quality-safe operating points tend to sit. It also breaks comparability: two systems reporting a nominal rate of 0.5 may be operating at different realised rates.

C. How much of the bias is correctable

A systematic error is only useful if it can be removed. Instances are split in half at random, a mean signed offset is fitted on one half either per budget or per budget and family, and the residual $|r - \text{target}|$ is measured on the held-out half. Table V reports 200 such splits.

A per-budget offset table, which needs no model and no per-instance computation, removes 54.7% of the error on unseen instances. Conditioning it on the task family removes 79.5%, cutting the residual by a further 55%. That is held-out evidence for the family dependence in Fig. 1: family identity carries information the requested budget does not.

The residual after per-family correction is 0.0084, which bounds from above what any per-instance adherence model could contribute. A two-column calibration table thus recovers most of the achievable correction, and a learned per-instance corrector has at most 0.008 of rate error left to work with on this compressor.

TABLE V

HELD-OUT MEAN $|r - \text{TARGET}|$ AFTER FITTING A SIGNED OFFSET ON A DISJOINT HALF OF THE INSTANCES, OVER 200 RANDOM SPLITS. BRACKETS ARE 2.5TH AND 97.5TH PERCENTILES ACROSS SPLITS.

Offset fitted on	error	95% interval	removed
nothing	0.0408	$[0.0380, 0.0435]$	n/a
budget	0.0185	$[0.0173, 0.0196]$	54.7%
budget $\times$ family	0.0084	$[0.0069, 0.0096]$	79.5%

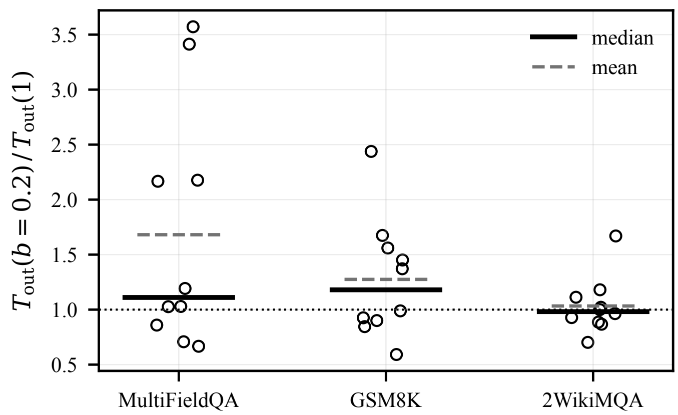

Fig. 2. Per-instance output expansion at $b = 0.2$ relative to $b = 1.0$. Each circle is one instance. The distribution is heavy-tailed on MultiFieldQA: the mean (dashed) is pulled well above the median (solid) by four instances, while three others contract.

D. Output expansion and the cost optimum

Compression lengthens model output on two of the three families. At $b = 0.2$ the mean ratio $T_{\text{out}}(0.2)/T_{\text{out}}(1)$ is 1.681 (95% CI $[1.268, 2.135]$) on MultiFieldQA and 1.275 ($[1.086, 1.481]$) on GSM8K, both excluding 1.0. On 2WikiMQA the ratio is 1.033 ($[0.889, 1.177]$), which does not exclude 1.0.

The mean is a poor summary here, and Fig. 2 shows why. On MultiFieldQA the ten per-instance ratios are heavy-tailed: the median is 1.110, four instances exceed 1.2 (reaching 2.17, 2.18, 3.41 and 3.57), and three instances contract to 0.67, 0.71 and 0.86. The effect is real in the mean but concentrated in a minority of instances at $n = 10$, and we treat the magnitude as exploratory rather than well estimated.

Whether expansion matters for cost depends on the magnitudes involved. On MultiFieldQA, moving from $b = 1.0$ to $b = 0.2$ reduces input from 2405 to 498 tokens, a saving of 1907, while output rises from 29.5 to 39.9 tokens, an increase of 10.4. Even at $\gamma = 5$ the comparison is 1907 against 52.

Applying the test of Section III-C directly: across all 25 instances with a non-empty safe set, at every price ratio $\gamma \in \{1, 2, 3, 4, 5\}$, the cheapest safe budget equalled the smallest safe budget. The divergence rate is 0 of 25 in every condition.

This is a bounded negative result, and it does not contradict the reported compression paradox [5]. That work measured

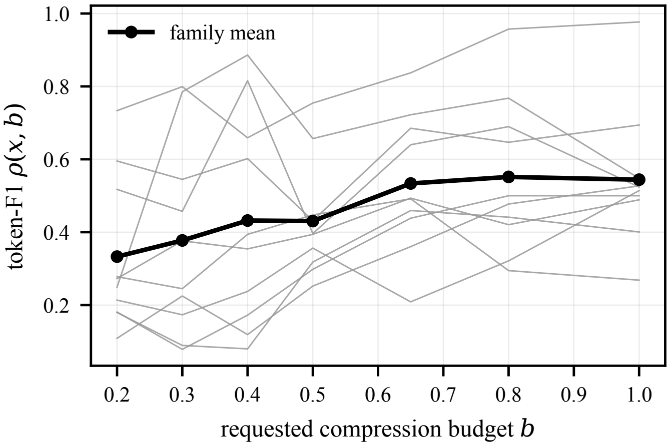

Fig. 3. Per-instance quality against requested budget on MultiFieldQA, the one family with no validity problems. Grey lines are individual instances, black is the family mean. The spread between instances is visible; Section III-C shows it is not separable from sampling noise at $k = 3$.

TABLE VI
HETEROGENEITY TEST ON COMPRESSION SENSITIVITY $S(x)$, RESTRICTED TO INSTANCES SOLVABLE WITHOUT COMPRESSION. NO FAMILY CLEARS THE CRITERION AT $k = 3$. REQUIRED k EXTRAPOLATES THE FLOOR AS $\propto 1/k$ AND IS AN ESTIMATE, NOT A MEASUREMENT.

Family	n	mean S	$\mathrm{Var}[S]$	floor	required k
MultiFieldQA	10	$+0.211$	0.0194	0.0655	20
2WikiMQA	9	$+0.049$	0.0022	0.0052	14
GSM8K	6	-0.056	0.1519	0.1690	7

code generation under truncation, where outputs are long relative to inputs and expansion factors reach 56×. The mechanism requires a high output-to-input token ratio. In short-answer question answering the ratio is roughly 1:80, and an expansion of even 1.68× cannot overcome it. The practical statement is conditional: optimising the smallest safe budget is adequate for short-output workloads, and the cheapest-safe rule is worth its extra complexity only when outputs are long relative to inputs.

E. Per-instance heterogeneity is not resolvable at $k = 3$

Fig. 3 shows the per-instance frontiers on MultiFieldQA, the only family that passes all validity preconditions (10 of 10 instances solvable uncompressed, $\rho(x, 1) = 0.544$). The instances visibly differ: one declines from 0.98 to 0.73 across the budget range while another falls from 0.53 to 0.11.

That visible spread is nevertheless not statistically separable from sampling noise. Pooled over families, $\mathrm{Var}[b^*] = 0.108$ with bootstrap CI $[0.074, 0.133]$ against a noise floor of 0.055, so the interval straddles the 2× floor criterion. Table VI repeats the test on the better-conditioned sensitivity statistic, and no family clears it.

The binding constraint is k, not n. The noise floor of a per-instance mean scales approximately as $1/k$, so the samples per cell needed for the observed between-instance variance to exceed twice the floor can be estimated. For MultiFieldQA that estimate is $k \geq 20$. This is an extrapolation from a floor measured at $k = 3$ and should be treated as a design estimate rather than a measured quantity.

The metric choice interacts with this. The exact-match floor on GSM8K is 0.1690 against 0.0655 for token-F1 on MultiFieldQA. Graded metrics carry roughly 2.5 times less per-instance sampling noise than binary ones at equal k, so a graded metric buys statistical power at no additional sampling cost.

F. Curve shape

Naive and suffix-safe definitions of b^* disagree on 20.0% of the 30 instances, with a mean gap of 0.483 in budget units. Aggregate quality peaks at $b = 0.8$ (0.428) rather than at $b = 1.0$ (0.372). We do not read the latter as compression improving accuracy; with $k = 3$ it is within the range that sampling noise produces, and it is one of the two conditions that make the pooled Gate-1 verdict uninterpretable. It does establish that the measured curves are not monotone, which is why a suffix-based safety definition is preferable to taking the smallest budget that happens to satisfy the tolerance.

VI. DISCUSSION

The three results differ in how much weight they can carry.

Rate adherence is well powered and consistent, resting on 1,044 measurements with no target-model calls, and every interval excludes zero. Its practical consequence is cheap to act on: a user targeting a quality-safe budget is served below it, and a two-column calibration table removes about four fifths of the discrepancy with no per-instance computation. The same measurement bounds the more elaborate alternative, since only 0.008 of rate error survives that table.

The cost result is a conditional negative. Expansion exists and is significant in the mean, but it does not move the optimum for this workload, and the reason is a token-ratio argument that transfers to other short-output tasks. Reporting the 1.68× expansion without the 0-of-25 divergence would reproduce, with the sign reversed, the reporting problem this work set out to examine.

The power result is the one that most changes what should be done next. The absence of a heterogeneity finding here is a statement about sample size, not about compression. Any study that estimates per-instance compression frontiers, then selects a rate from them, needs enough samples per cell for the per-instance label to be meaningful, and $k = 5$ is a common choice that our measurement suggests is too small for graded QA metrics by roughly a factor of four.

A. The experiment that would settle the open question

The power analysis specifies its own follow-up: holding the instance count at ten and raising the sample count to $k = 20$ on MultiFieldQA gives 1,400 cells, about three hours at the 7 s per cell measured here. Either outcome is informative. If the between-instance variance then exceeds twice the recomputed floor, per-instance rate selection has a measured basis; if it does not, the result is a well-powered negative and a per-family fixed budget is the appropriate policy. The present study cannot distinguish these cases.

VII. Limitations

The scope of every claim is one target model, one compressor checkpoint, and one context-length window.

Single model. All quality and output-length measurements use Llama-3.2-3B-Instruct at 4-bit quantisation. Output-length response to compression is known to be model-dependent [13], so the expansion direction should not be generalised.

Single compressor. Only LLMLingua-2 with the multilingual BERT-base checkpoint was run. Adherence behaviour is a property of the compressor and may not transfer. The checkpoint is trained on MeetingBank; none of our three families is MeetingBank, so the results are out of domain for the compressor, which we regard as the appropriate default.

Sample size. Thirty instances at $k = 3$ supports the adherence result, which does not depend on generation, but is underpowered for heterogeneity and gives wide intervals on expansion. The expansion magnitude in particular rests on ten instances with a heavy-tailed distribution.

Task coverage. As Table III shows, quality conclusions rest on MultiFieldQA alone, so they are supported by ten instances. Excluding HotpotQA and QMSum for length means long-context behaviour, which is where compression is most often deployed, is untested.

Tokenizer proxy. Realised rate uses a GPT-2 tokenizer rather than the target's, which shifts the absolute level of the adherence error.

Cost model. Local inference is zero-priced, so all cost statements are in token units under an assumed price ratio, and exclude compressor compute. Kummer et al. [4] show that compressor overhead can cancel end-to-end gains outside a matched operating window; our token-unit accounting does not capture that.

Runtime dependence. Two measurements affect reproduction rather than interpretation. Running the compressor on the Apple GPU backend was about 17 times faster than CPU (0.54 s against 9.42 s mean per compression), but GPU residency of the compressor alongside the target model caused memory pressure that terminated an earlier run, so the reported grid used the CPU backend. Separately, issuing generation requests concurrently reduced throughput on this single-GPU machine (12.0 s per cell at one worker against 16.8 s at four) because concurrent prefills contend for the same device; prefill-bound workloads of this kind do not benefit from request-level parallelism.

No method. No predictor, selection policy or risk guarantee was built or evaluated. Statements about what a learned adherence correction would buy are hypotheses.

VIII. Conclusion

We measured what a requested compression rate delivers. For LLMLingua-2 on a 3B target model, the realised rate is consistently below the requested rate, by an amount that varies non-monotonically with the budget and by a factor of about three across task families. The compressor is more aggressive than asked, so the realised saving exceeds the requested one while the delivered quality corresponds to a budget the user did not select. A calibration table indexed by budget and family removes 79.5% of the discrepancy on held-out instances, leaving 0.008 of rate error as the ceiling on what a per-instance model could recover. Compression lengthens output on two of three families, significantly so in the mean, but the increase never changed the cost-optimal budget in our data, because input reduction dominates output growth by about two orders of magnitude in short-answer tasks. The smallest-safe-budget rule is adequate for such workloads; the distinction matters only where outputs are long relative to inputs.

We could not establish per-instance heterogeneity in compression tolerance. The curves show visible spread, but at three samples per cell it is not separable from sampling noise, and our estimate is that roughly twenty samples per cell would be needed for graded QA metrics. That is the main obstacle to per-instance rate selection that this study identifies, and addressing it is a precondition for the predictor and calibration work that motivated the measurement.

Reproducibility

The ledger, the analysis scripts and the figure-generation code are released with the paper. Every number reported here is regenerated from `runs/real_v1/ledger.jsonl` and `runs/adherence_v1/adherence.csv` by the included scripts. The test suite accompanying the harness passes 136 tests with 5 skipped.

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